Recall

Isom Thm If $\dim_{\mathbb{F}} V = n < \infty$ Then $V \cong \mathbb{F}^n$

EX. $V = \{y \mid y'' + 2y' - 3y = 0\}$

$\dim_{\mathbb{R}} V = 2 \Rightarrow V \cong \mathbb{F}^2$ (basis: e^x, e^{-3x})

FTofLA $\mathrm{Hom}_{\mathbb{F}}(\overset{n}{V}, \overset{m}{W}) \overset{\Phi}{\cong} \mathbb{F}^{m\times n}$

EX. $W = sl_2(\mathbb{F}) = \{A \in \mathbb{F}^{2\times 2} \mid \mathrm{tr}\, A = 0\}$. Find $V \xrightarrow[A_{3\times 2}]{\sigma} W$

$\dim W = 3$ (basis: $E_{11} - E_{22}, E_{12}, E_{21}$)

$A_{3\times 2}: (e^{-3x}, e^x)\alpha \longmapsto (w_1, w_2, w_3)(A\alpha)$

Q 1. $V + W = \{v + w \mid v \in V, w \in W\}$. × (不一定能进行)

2. $V \times W = (\{(v, w) \mid v \in V, w \in W\}, +, \cdot)$

$v + w$ meaningful $\xrightarrow{\text{Forcing}}$ $V, W \subseteq X$ (subspace)

$V + W = \{v + w \mid v \in V, w \in W\}$ $\quad \mathcal{L}(V) = \{U \subseteq V \mid U \text{ is a subspace of } V\}$.

$\forall V, W \in \mathcal{L}(X)$

- $V + W \overset{\sigma}{\hookrightarrow} V \times W$ is an injection $\Rightarrow \mathrm{Im}\,\sigma \cong V + W$

$v + w \longmapsto \sigma(v + w) = (v, w)$ → This is not well-defined

Moreover, σ is an isomorphism $\Leftrightarrow V \cap W = 0$ i.e. $V + W = V \oplus W$

- Q: $\sum_{i=1}^{\infty} \oplus V_i \ncong V_1 \times V_2 \times \cdots \times V_n \times \cdots (= \prod_{i=1}^{\infty} V_i)$ $\quad \forall V_i \subseteq \mathcal{L}(X)$

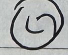
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad$ ↑ product

$\parallel \{v_{i1} + v_{i2} + \cdots + v_{ik} \mid v_{ij} \in V_{ij}, k \in \mathbb{N}^+\}$

$\coprod_{i=1}^{\infty} V_i$ ← coproduct

Quotient

Ex. S^1 ○ $\times$ S^1 ○ $\neq S^2$ $\qquad S^1 = \text{Circle} \qquad S^n = \{(x_1, x_2, \cdots x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1\}$

$\{(x_1, x_2) \mid x_i \in S^1\} =$  $\qquad S^2 = \text{sphere}$

Partition 划分 $\phi \neq S = S_1 \dot\cup S_2 \dot\cup \cdots \dot\cup S_k$

$S_1 \dot\cup S_2 = S_1 \cup S_2$ and $S_1 \cap S_2 = \phi$ (disjoint)

i.e. $S = \dot\bigcup_{i=1}^{k} S_i$, $S_i \cap S_j = \phi$ whenever $i \neq j$

Define equivalence relation (等价关系) on S

by $x \sim y \Leftrightarrow x, y \in S_i$, $i = 1, 2, \cdots k$

Ex. $\mathbb{Z} = \underset{=[0],\bar{0}}{2\mathbb{Z}} \cup \underset{=[1],\bar{1}}{(2\mathbb{Z}+1)}$ $\qquad \mathbb{Z}_2 = (\{\bar{0}, \bar{1}\}, +, \cdot)$

Let $U, V \in \mathcal{L}(W)$, $U \oplus V = W$

Try to define W/U

Ex. $\mathbb{R}^1 \times \mathbb{R}^1 = \mathbb{R}^2$ $\quad (\mathbb{R}^1 \times 0 \oplus 0 \times \mathbb{R}^1 = \mathbb{R}^2)$

Lemma Let $U \in \mathcal{L}(W)$ Then

$\alpha \sim \beta \Leftrightarrow \alpha - \beta \in U$ is an equivalence relation

moreover, $\alpha \sim \beta \Leftrightarrow \beta \in \alpha + U = \{\alpha + u \mid \forall u \in U\}$

THM (Construction of Quotient space)

Let $U \in \mathcal{L}(W)$ Denote by $W/U = \{\bar{w} \mid w \in W\}$, where $\underset{\text{coset 陪集}}{\bar{w} = w + U}$

Then $(W/U, +, \cdot)$ is a v.s. $/\mathbb{F}$,

where $(w + U) \overset{\text{“}+\text{”}}{+} (w' + U) = (w + w') + U$ $\qquad \bar{0} = 0 + U = U$

$\text{“}\cdot\text{”} \ \lambda(w + U) = \lambda w + U$ $\qquad -(v + U) = -v + U$

$[w + U = v + u + U = v + U \Rightarrow W/U \cong V]$